HARBOURSIDE CANOPY by B+P ARCHITECTS

HSINCHU, TAIWAN - 2025

DESCRIPTION OF THE PROJECT

Harbourside Canopy by B+P Architects is a lightweight, open-air timber and steel canopy that shelters a 496 m² fishing market pavilion on the waterfront in Hsinchu, Taiwan. The canopy structure spans the working zone, linking the dock-side fishing activity with the inland marketplace and providing shelter from wind and rain while framing the rhythm of everyday harbor life.



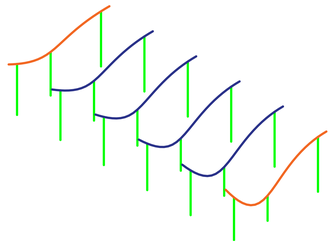
I. CANOPY GEOMETRY

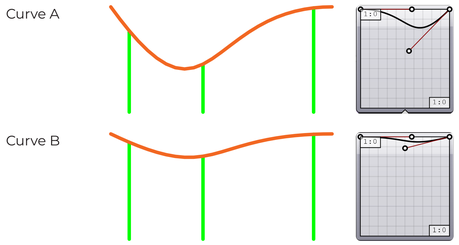
The beam geometry is defined by two Bézier curves located at both ends of the roof. A series of four additional beams is generated by interpolating and connecting these two primary curves.

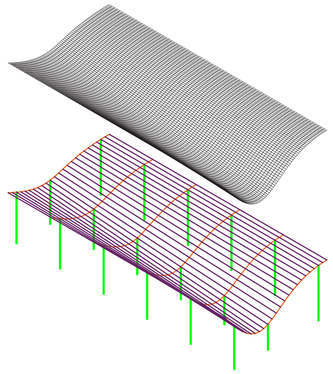
Each beam is supported by three columns: two located approximately at the beam ends and one positioned near the boundary of the middle third of the span.

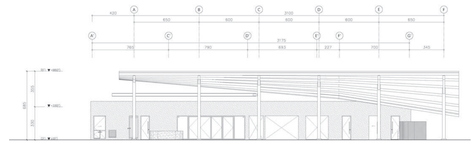
Twenty-four secondary beams (timber joists) are arranged above the primary curved beams at a spacing of 600 mm. The roof finish sits directly on top of the joists, following the curvature of the steel beams. To accurately represent this geometry, the roof surface was meshed, and the resulting mesh geometry was imported into MIDAS for structural analysis.

These structural sections were adopted for the primary beams, columns, and timber joists of the canopy system.

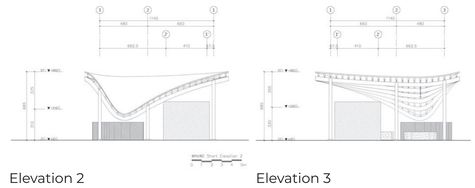








Elevation 1



Elevation 2 Elevation 3

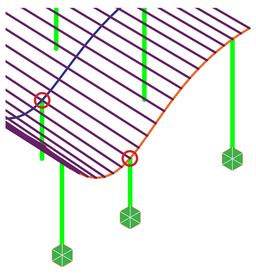
Primary Beams
RHS 300X250X10 - S355

Columns
CHS Ø323.9X12.5 - S355

Timber Joists
250X150 - C24

II. SUPPORTS + BOUNDARIES

Column base connections and welded connections were modelled as rigid joints in MIDAS providing full restraint of all translational and rotational motions. (Dx, Dy, Dz, Rx, Ry, Rz, Rw).

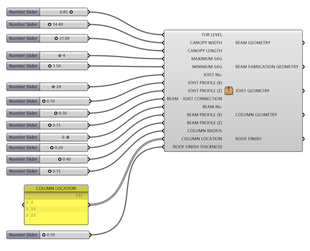


For the purpose of the finite element analysis, the structural model has been done such that all primary members (beams, columns, joists, and the roof mesh) converge at a single node. This approach simplifies the connectivity in the FE model, ensuring load transfer between members is fully captured while reducing computational complexity.

III. PARAMETERS & 3D GEOMETRY DEFINITION

Grasshopper and Rhino

All the Beam Curves have been approximated to line segments and the roof was converted into a Mesh before exporting the .dxf file to Midas.



IV. LOADS

1. DEAD LOADS

STRUCTURE SELF-WEIGHT

Calculated by Midas based on the Model Properties and Materials.

ROOF FINISHING AND MATERIALS LOAD

Set to **0.50kN/m2**

2. LIVE LOADS

0.25kN/m2 Maintenance only, no public access.

Note: All the Loads were assigned to the Roof mesh Geometry as a Pressure Loads in Global X,Y,Z Directions

3. WIND LOADS:

Mean Wind Velocity $v_m(z)$

Taiwan is typhoon-prone so the Mean Wind Velocity is considered to be around **40 m/s.**

Peak Wind Velocity $q_p(z)$

$$q_p(z) = \frac{1}{2}\rho v_m^2(z)\,[1 + 7I_v(z)] \quad \text{OR} \quad q_p(z) = 0.613 \cdot v_m^2$$

$$q_p = 0.613 \cdot 40^2 = 0.613 \cdot 1600 = \boxed{0.98\ \text{kN/m}^2}$$

Uplift Pressure w_z

An approximate pressure coefficient of **-1.5** was adopted to account for the combined effects of edge zones and the mean roof surface.

$$w_z = q_p \cdot c_{pe} \quad w_z = 0.98 \cdot (-1.5) = \boxed{-1.47\ \text{kN/m}^2}$$

Horizontal Wind Forces $w_x\ w_y$

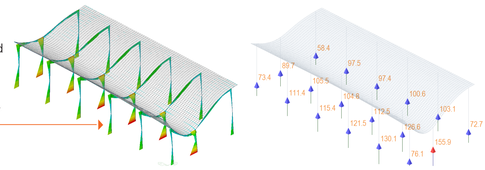
Horizontal wind actions were evaluated using a force coefficient of **1.2** appropriate for an open-sided canopy.

$$w_x = w_y = q_p \cdot c_f = 0.98 \cdot 1.2 = \boxed{1.18\ \text{kN/m}^2}$$

V. ANALYSIS GOAL

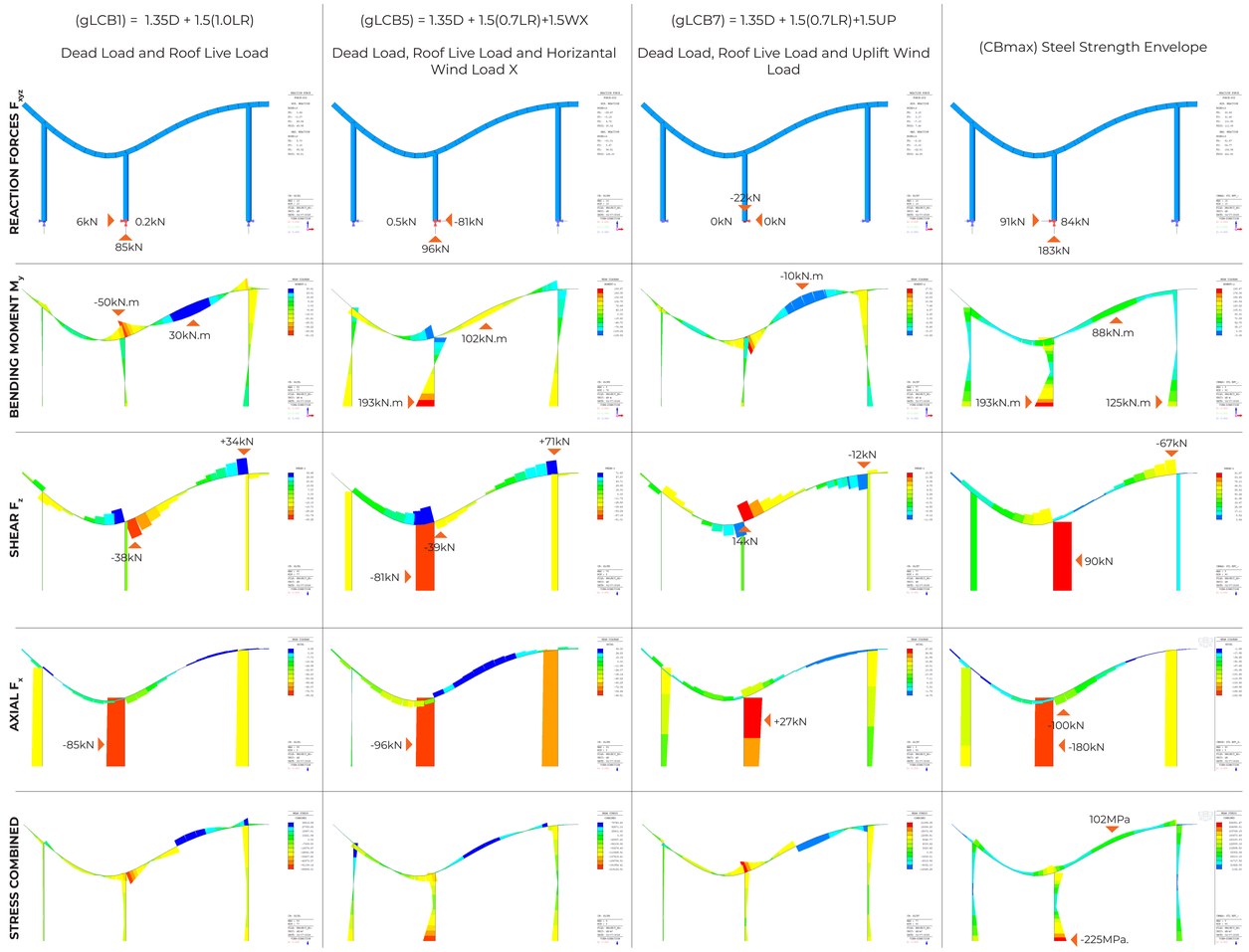
An investigation was carried out to assess the effect of wind actions on the canopy by comparing structural responses with and without wind load cases; the results show a significant influence of wind due to the high wind intensity associated with the site location in Taiwan.

The structural frame analyzed and presented in the results comprises one steel beam supported by three steel columns.



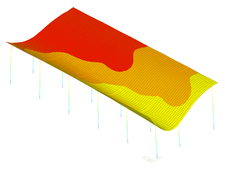
Note: Timber joists are excluded from detailed analysis, as their internal forces are minimal relative to the steel beam and columns, and therefore do not significantly influence the structural response.

VI. RESULTS ULS LOAD COMBINATIONS

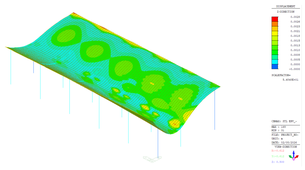


VII. DEFORMATIONS SLS LOAD COMBINATIONS

(CBmax) Steel Serviceability Envelope DXYZ



(CBmax) Steel Serviceability Envelope DZ



Lateral Displacement in Y Direction (Δmax = 3cm)

Δ lim= 1/300 L = 1/300 · 30m = 10cm

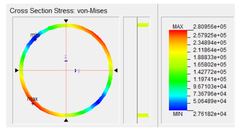
Δmax / Δ lim = 30% Passed

Displacement in Z Direction at Mid Span (Δmax = 1.18cm)

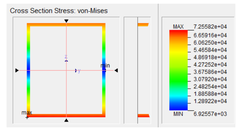
Δ lim= 1/300 L = 1/300 · 7.2m =2.4cm

Δmax / Δ lim = 49% Passed

VIII. CROSS SECTION STRESS: VON-MISES



CHS 323.9×12.5
Max Stress 280MPa



RHS 300×250×10
Max 72MPa

CHS 323.9×12.5
280 / 308 = 90%
Passed

RHS 300×250×10
72 / 308 = 23.3%
Passed

IX. VERIFICATIONS

DESIGN RESISTANCES WITH PARTIAL FACTOR Y_{M0} = 1.15

Resistance	CHS 323.9×12.5	RHS 300×250×10
Axial(N)	3,775 kN	3,270 kN
Bending (max)	358 kNm	448 kNm
Shear (V)	2,180 kN	1,900 kN
Stress	308 MPa	308 MPa

MAX VALUES

Resistance	CHS 323.9×12.5	RHS 300×250×10
Axial(N)	- 180kN	- 100kN
Bending (max)	193kNm	102kNm
Shear (V)	90kN	71kN
Stress	225 MPa	102 MPa

% OF DESIGN RESISTANCES

Resistance	CHS 323.9×12.5	RHS 300×250×10
Axial(N)	4.75% Passed	3.05% Passed
Bending (max)	53.9% Passed	22.7% Passed
Shear (V)	4.12% Passed	3.73% Passed
Stress	73.0% Passed	33.1% Passed

X. CONCLUSION

The analysis shows that horizontal wind loads govern the structural response, producing the majority of the internal actions. These actions are primarily bending moments in both columns and beams, which are efficiently transformed into stresses in the steel members.

Axial and shear forces are comparatively small, with columns carrying most of the axial load and beams acting mainly in bending and shear. All stresses, including combined axial and bending effects, have been checked against the steel design limits, and the structure is verified to be safe and adequate under the applied load cases.

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061759 - METHODS AND TECHNIQUES FOR MODELING STRUCTURES (2025-26)
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